What’s Wrong With Science?

Session 1: What is science? The history and philosophy of science.

Professor David Gresham
Fall 2026

What makes something science?

1. “People who are born with Mars as their ruling planet are more aggressive”

2. “People who inherit two copies of a particular variant of the hemoglobin gene will develop sickle cell anemia”

What makes something science?

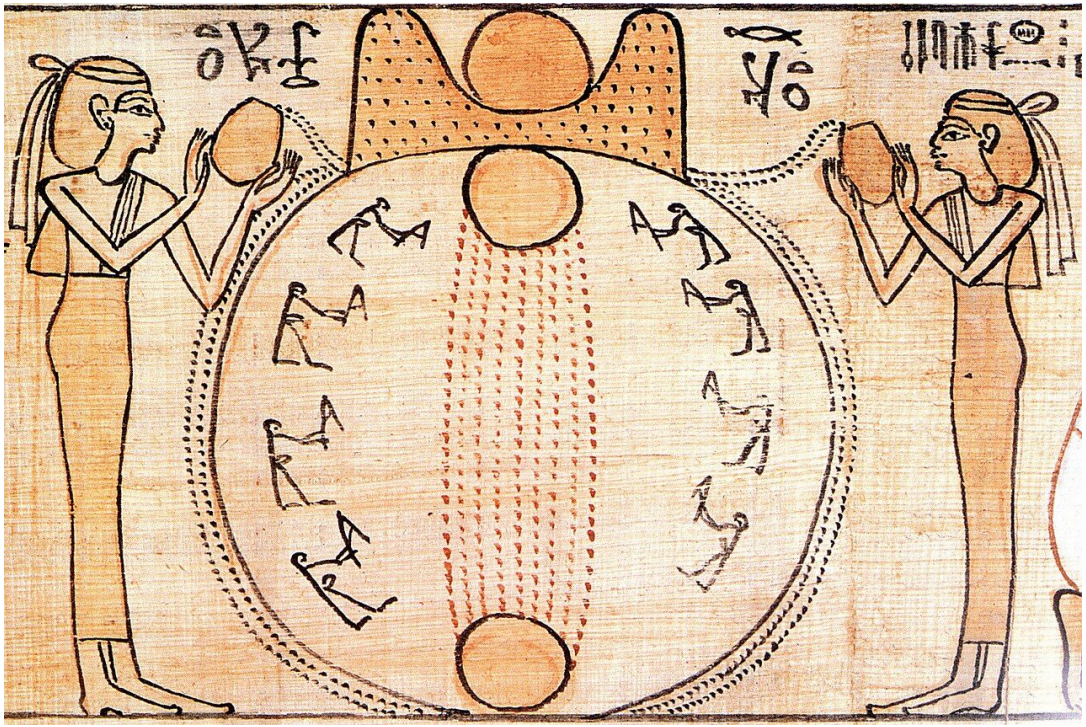
1. “People who are born with Mars as their ruling planet are more aggressive” - Astrology

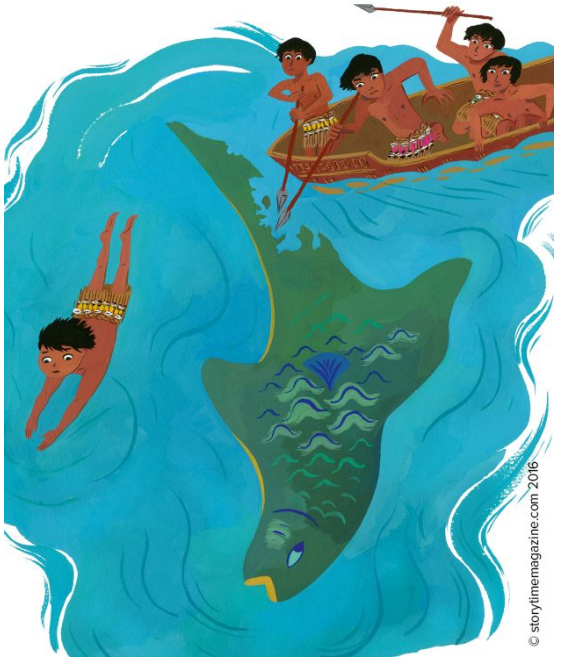
2. “People who inherit two copies of a particular variant of the hemoglobin gene will develop sickle cell anemia” - Genetics

- Both statements make predictions about people based on conditions at birth
- Both astrology and genetics have practitioners, journals, and technical vocabularies.
- Astrology has been around a lot longer

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

For millennia humans explained the world through creation myths





Greek Natural Philosophy - science(ish)



Aristotle (384 to 322 BCE)

- **Systematic empirical observation** — insisted that knowledge of nature must begin with careful observation of actual specimens rather than pure reasoning, setting the template for descriptive natural history.
- **Biological classification** — described roughly 500 animal species and grouped them by shared characteristics (blooded vs. bloodless, live-bearing vs. egg-laying), an early hierarchical taxonomy that anticipated Linnaeus by two millennia.
- **Embryology** — opened chicken eggs at daily intervals to document development, correctly describing the beating heart appearing early and the sequential formation of organs (epigenesis rather than preformation).
-
- **Matter** - made up of earth, air, fire, and water
- **Spontaneous generation** — held that insects, eels, and various lower animals arise directly from mud, dew, or rotting matter; this persisted until Redi and finally Pasteur dismantled it.
- **The heart is the seat of intelligence and the brain is a radiator** — believed the brain's function was to cool the blood, and that women have fewer teeth than men (apparently without ever counting).

Medieval Europe: Fusing of Aristotle and Christian Doctrine

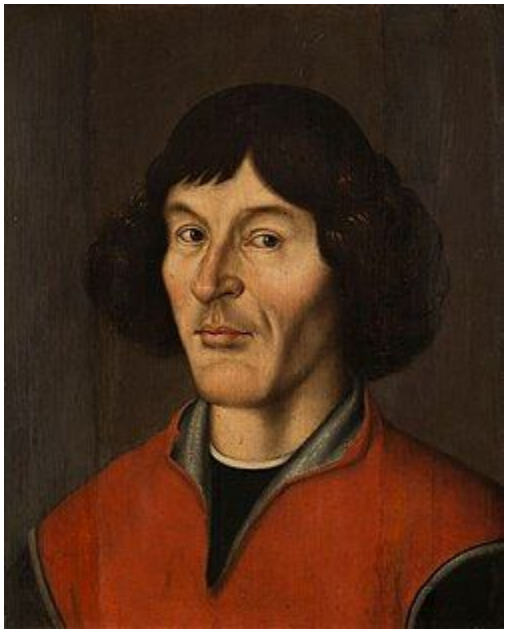


University of Bologna AD 1088



- Universities trained scholars to interpret authoritative texts
- The question was not “what does nature show?” but “what did Aristotle or Scripture mean?”

The Scientific Revolution, (1543–1687)



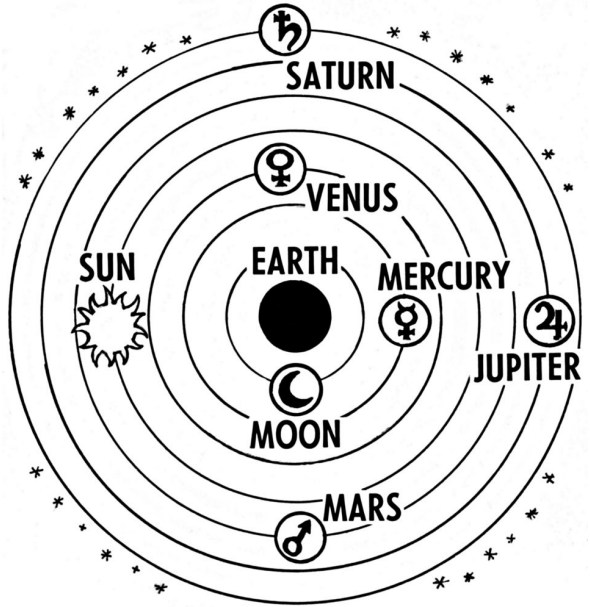
Nicolaus Copernicus (1473 - 1543)

NICOLAI CO
PERNICI TORINENSIS
DE REVOLVTIONIBVS ORBI
um cœleſtium, Libri VI.

Habes in hoc opere iam recens nato, & ædito, ſtudioſe lector, Motus ſtellarum, tam fixarum, quàm erraticarum, cum ex ueteribus, tum etiam ex recentibus obſeruationibus reſtitutos: & nouis inſuper ac admirabilibus hypotheſibus ornatos. Habes etiam Tabulas expeditiſsimas, ex quibus eoſdem ad quoduis tempus quàm facilli me calculare poteris. Igitur eme, lege, fruere.

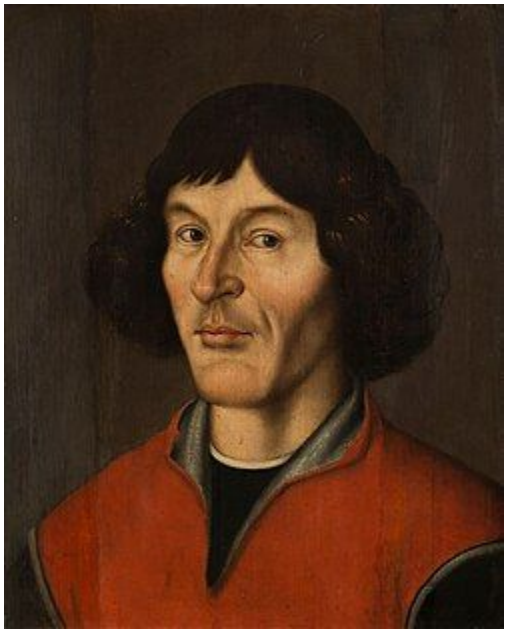
Ἀγεωμέτρητος μηδεὶς εἰσίτω.

Norimbergæ apud Ioh. Petreium,
Anno M. D. XLIII.



Geocentric
(Ptolemy)

The Scientific Revolution, (1543–1687)



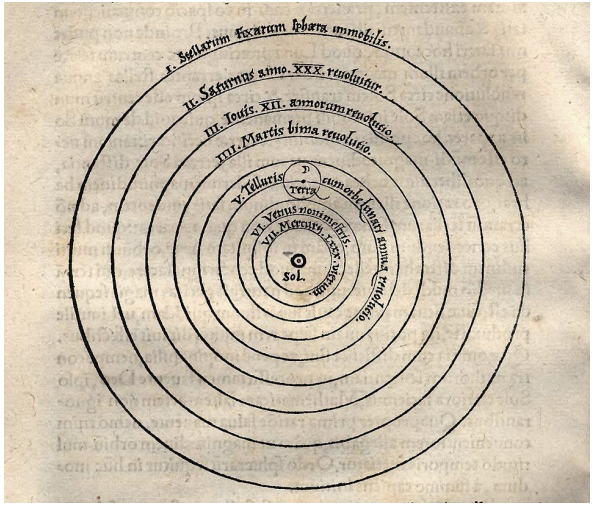
Nicolaus Copernicus (1473 - 1543)

NICOLAI CO
PERNICI TORINENSIS
DE REVOLVTIONIBVS ORBI-
um cœleſtium, Libri VI.

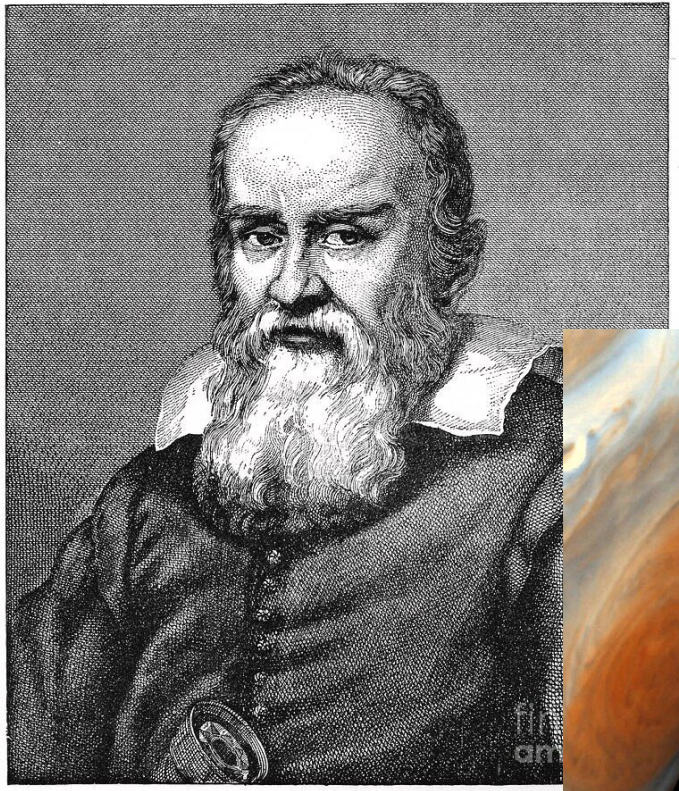
Habes in hoc opere iam recens nato, & ædito, ſtudioſe lector, Motus ſtellarum, tam fixarum, quàm erraticarum, cum ex ueteribus, tum etiam ex recentibus obſeruationibus reſtitutos: & nouis inſuper ac admirabilibus hypotheſibus ornatos. Habes etiam Tabulas expeditiſsimas, ex quibus eoſdem ad quoduis tempus quàm facilli me calculare poteris. Igitur eme, lege, fruere.

Ἀγεωμέτρητος μηδεὶς εἰσίτω.

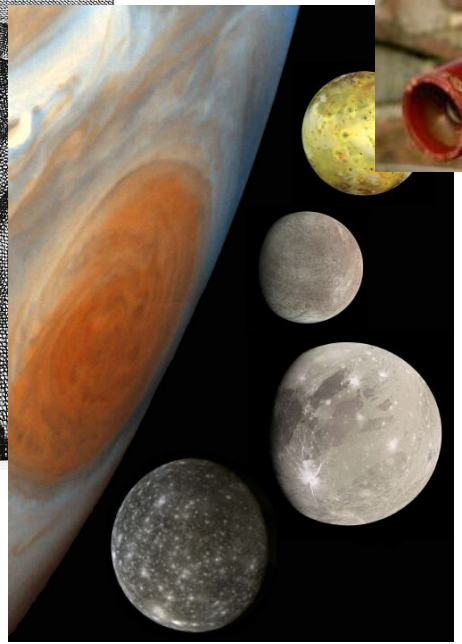
Norimbergæ apud Ioh. Petreium,
Anno M. D. XLIII.



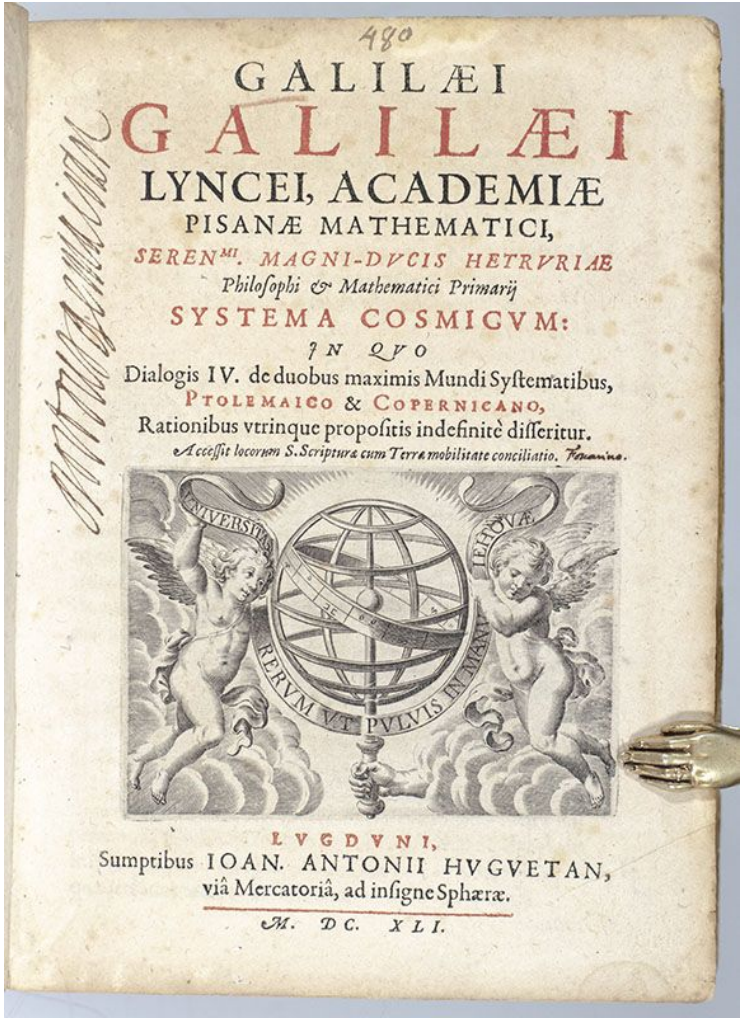
Heliocentric







Galileo Galilei (1564 - 1642)



Pope Urban VIII gave Galileo permission to write a book comparing the Earth-centered and Sun-centered models—**under the strict condition** that it remain unbiased and present heliocentrism *only* as a mathematical hypothesis, not absolute truth

***Dialogue Concerning the Two Chief World Systems* (1632)**. Instead of being neutral, the book was a scathing defense of heliocentrism. Worse, it was written as a conversation between three characters:

1. A brilliant pro-Copernican philosopher (essentially Galileo).
2. An open-minded observer.
3. **Simplicio** (meaning "Simpleton"), a slow-witted defender of the geocentric view.



In 1633, the Roman Inquisition placed scientist Galileo Galilei on trial for promoting heliocentrism—the scientific model stating that the Earth revolves around the Sun. Found **"vehemently suspect of heresy,"** Galileo was forced to recant his findings and spend the remaining nine years of his life under **house arrest**



Francis Bacon (1561 - 1626)

The “father of empiricism” and the scientific method (published Novum Organum (1620))

Empiricism over authority — knowledge from observation and experiment, not passive observation of nature or received doctrine
Critique of the Idols of the Mind — defined biases that distort human reasoning
Inductive method — building general conclusions gradually from accumulated particular observations, rather than deducing from fixed axioms
Knowledge as power — science's purpose is practical utility and human benefit, achieved through organized, collaborative inquiry

The Royal Society (founded 1660)

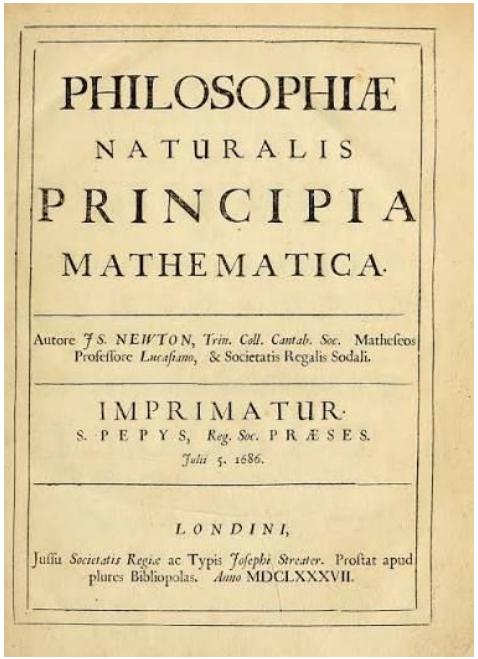
College for the Promoting of Physico-Mathematicall Experimentall Learning



Royal Society Motto: *Nullius in verba* (take nobody's word for it)



Isaac Newton (1643–1727)

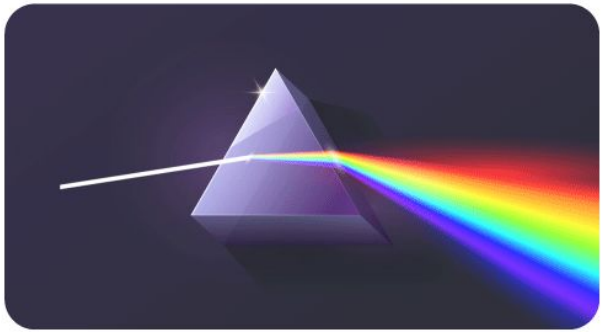


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore *J S. NEWTON, Trin. Coll. Cantab. Soc.* Matheseos Professore *Lucasiano*, & Societatis Regalis Sodali.

IMPRIMATUR.
S. PEPYS, *Reg. Soc.* PRÆSES.
Julii 5. 1686.

LONDINI,
Jussu *Societatis Regiæ* ac Typis *Josephi Streater*. Prostat apud plures Bibliopolas. *Anno* MDCLXXXVII.



- **Published 5 July 1687**
- **The laws of motion**
- **Universal Gravitation**
- **Calculus Foundations**

The The Scientific Revolution, (1543–1687): From Copernicus to Newton

1543

Copernicus

De Revolutionibus — the Sun, not Earth, at the center

1610

Galileo

Turns a telescope on the heavens

1620

Bacon

Novum Organum — knowledge from induction, not authority

1633

Galileo's Trial

The Church condemns heliocentrism

1687

Newton

Principia — one law, unifying heaven and Earth

What was new?

- Empiricism: knowledge built up from careful observation, not first principles
- Mechanism : nature as governed by law-like, describable regularities
- New instruments extend the senses: telescope, microscope
- Mathematics becomes the language for describing nature precisely
- Observation begins to outrank inherited authority

What Counts as Evidence?

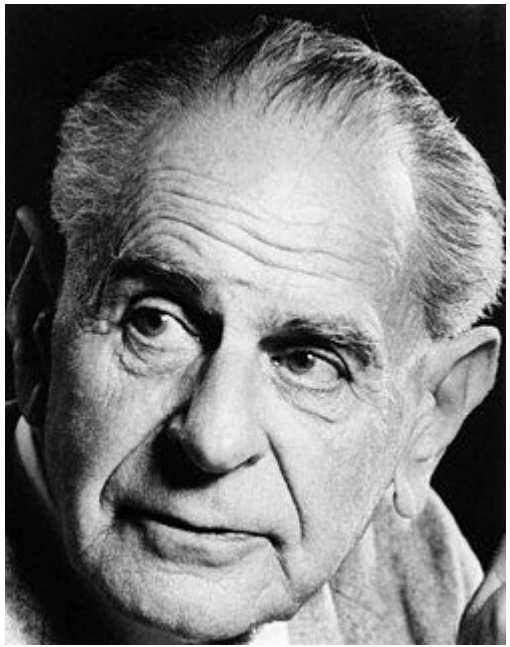
- Many of Galileo's contemporaries refused to even look through his telescope
- Was an instrument-mediated image trustworthy, or an optical illusion?
- No agreed standard yet existed for validating a new kind of observation
- Science had to invent its own credibility — replication, shared instruments, public demonstration

Science Needed Institutions

- Royal Society of London (1660) and Académie des Sciences (1666) founded
- Motto of the Royal Society: Nullius in verba — “take no one's word for it”
- Experiments performed before witnesses; results published for scrutiny
- Peer witnessing and publication become the new form of authority

What Is Science, and How Does It Work?

Popper: Falsifiability



Karl Popper (1902 - 1994)

- Karl Popper (1902–1994) proposed falsifiability as the line between science and non-science
- A theory is scientific only if it makes risky predictions — ones that could, in principle, prove it wrong
- Astrology and freudian psychoanalysis can explain almost any outcome after the fact — unfalsifiable
- Science advances by conjecture and refutation: propose bold theories, then try hard to prove them false
- A theory gains credibility not by being confirmed, but by surviving serious attempts at refutation

For example, “All swans are white”



For example, “All swans are white”







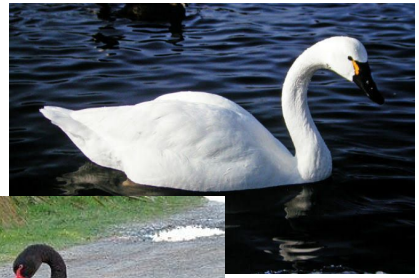


For example, “~~All swans are white~~”











Non-falsifiable statements

Non-falsifiable statements

- Everything that happens, happens for a reason.

Non-falsifiable statements

- Everything that happens, happens for a reason.
- There is an undetectable teapot orbiting the Sun

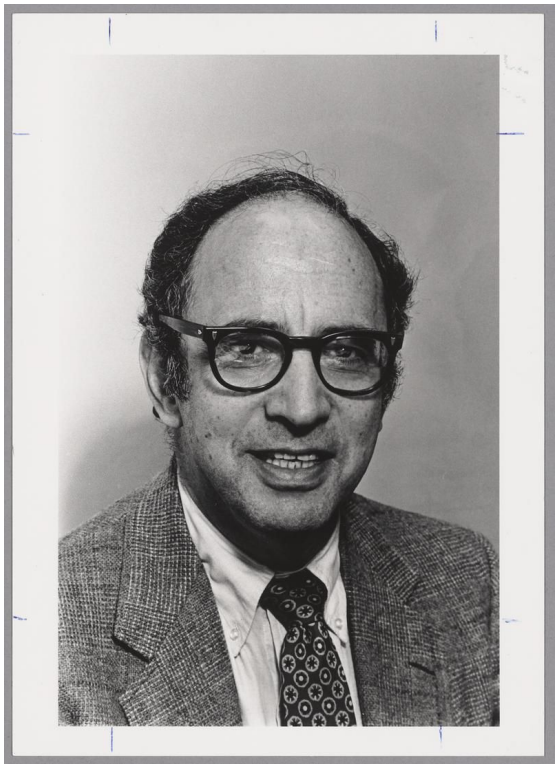
Non-falsifiable statements

- Everything that happens, happens for a reason.
- There is an undetectable teapot orbiting the Sun
- The universe was created five minutes ago complete with false memories and fossils

Non-falsifiable statements

- Everything that happens, happens for a reason.
- There is an undetectable teapot orbiting the Sun
- The universe was created five minutes ago complete with false memories and fossils,
- This trait exists because it was adaptive at some point in the past

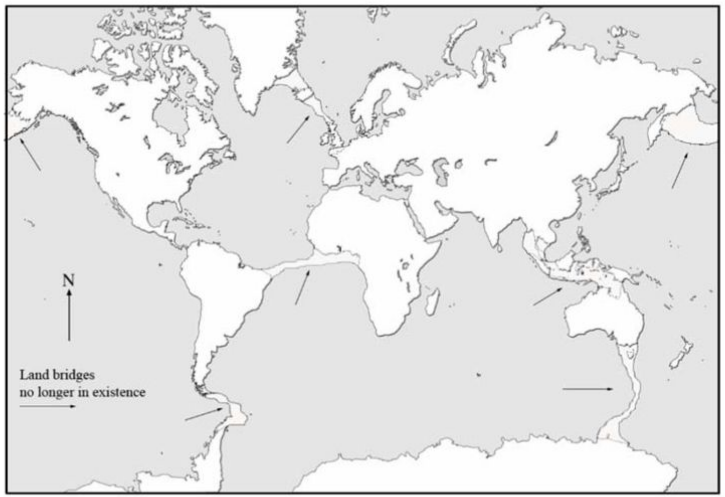
Kuhn: Paradigms and Revolutions



- Thomas Kuhn (1922–1996), The Structure of Scientific Revolutions (1962)
- “Normal science”: researchers work inside a shared paradigm, solving the puzzles it defines
- Paradigm = which problems count as legitimate, which methods are acceptable, what the key terms mean, and what counts as a good solution
- Anomalies accumulate over time until the paradigm enters crisis and a scientific revolution replaces one paradigm with another
- Paradigms are “incommensurable”: successive paradigms can't be fully compared on shared terms — scientists don't just change their minds, they change how they see things

What explains the occurrence of similar fossils on different continents?

Land Bridges



What explains the occurrence of similar fossils on different continents?

Land Bridges

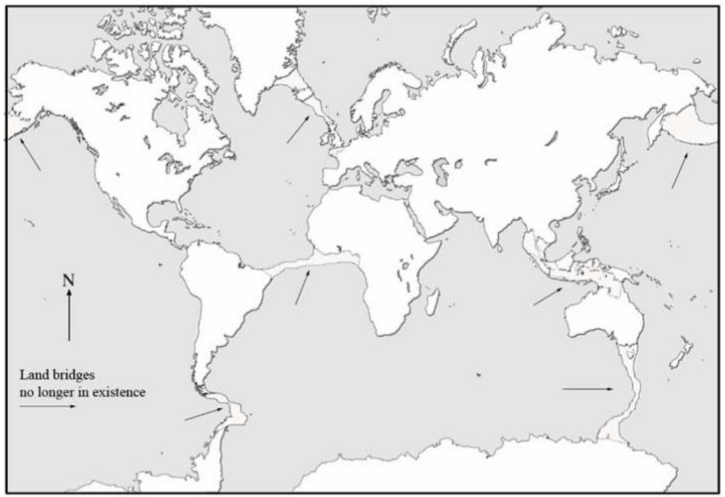
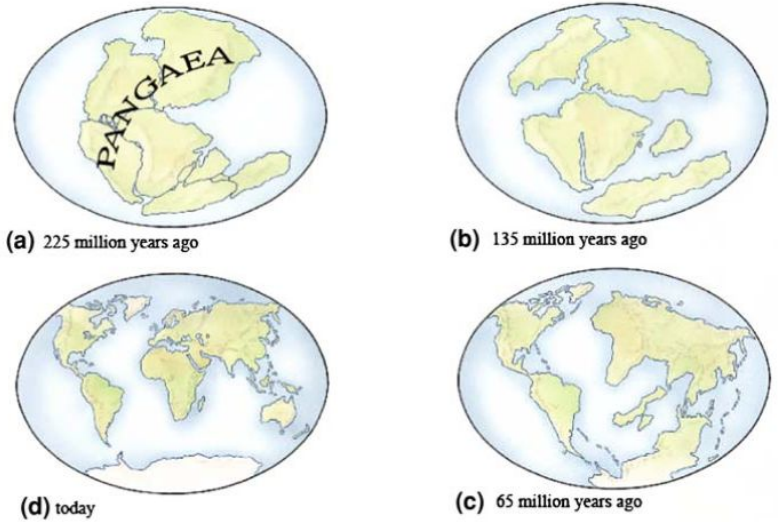


Plate tectonics



Is It Science?

1. **Astrology** — interpreting the positions of planets and stars to describe personality and predict events.
2. **Climate modeling** — using computer simulations of the atmosphere and oceans to project future climate.
3. **Cognitive therapy** — a form of psychotherapy that treats problems by changing patterns of thought.
4. **Economics** — the study of how people, firms, and governments produce, trade, and consume resources.
5. **Evolutionary psychology** — explaining human behavior and mental traits as adaptations shaped by natural selection.
6. **Forensic bite-mark analysis** — matching bite marks on a victim to a suspect's teeth for use in court.
7. **Homeopathy** — treating illness with highly diluted preparations of substances that cause similar symptoms.
8. **Machine-learning medical diagnosis** — algorithms trained on patient data or images to detect disease.
9. **Nutrition epidemiology** — studying links between what populations eat and the diseases they develop.
10. **String theory** — a framework in physics proposing that fundamental particles are tiny vibrating strings.
11. **Traditional ecological knowledge** — knowledge of local ecosystems accumulated by Indigenous and local communities over generations.
12. **Vaccine safety monitoring** — tracking adverse events after vaccines are licensed to detect rare side effects.

Is it science?

(A) astrology

0%

(B) evolutionary psychology

0%

(C) nutrition epidemiology

0%

(D) forensic bite-mark analysis

0%

(E) string theory

0%

(F) climate modeling

0%

(G) homeopathy

0%

(H) machine-learning medical diagnosis

0%

(I) economics

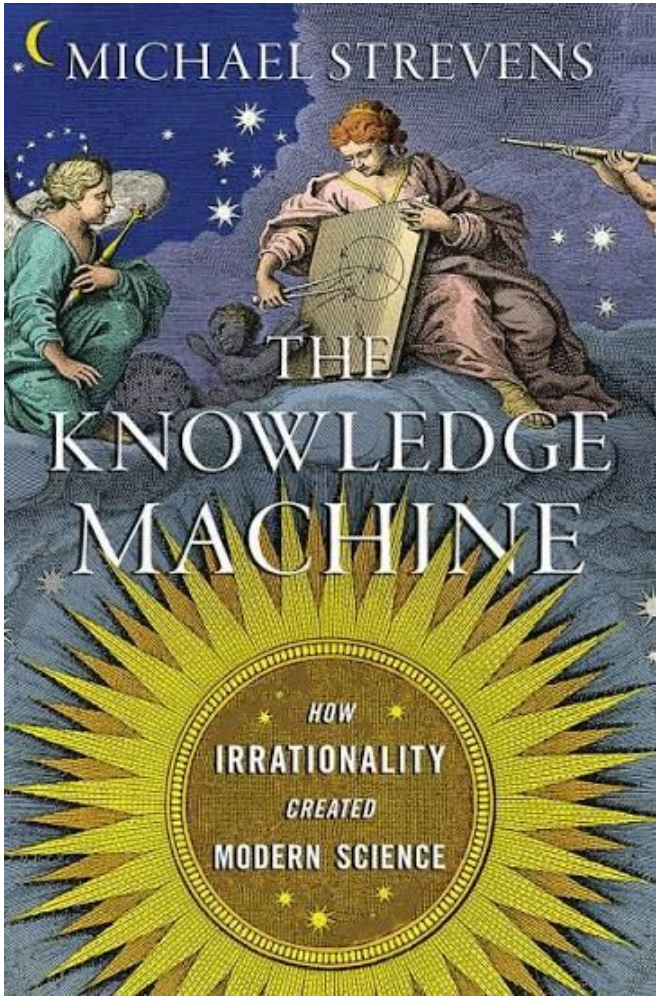
0

(J) cognitive therapy

0

SEE MORE

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**


MICHAEL STREVENS
THE KNOWLEDGE MACHINE
HOW IRRATIONALITY CREATED MODERN SCIENCE